

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Analysed customer behaviour patterns and transaction data across **PostgreSQL** and **Oracle** databases to identify retention opportunities, segmentation strengths and weaknesses, and priority customer cohorts for targeted **CRM and loyalty** campaigns, delivering actionable insights to cross-functional business teams on category-level customer value growth and reactivation potential.
- Designed and built **Power BI** dashboards and **Tableau** interactive reports tracking **CRM and loyalty performance KPIs**, campaign uplifts, and **return on investment** across incentive and voucher programmes, enabling real-time visibility into customer acquisition, retention metrics, and segment-level performance against business goals.
- Performed **advanced customer segmentation and cohort analysis** using **Python** (pandas, NumPy, scikit-learn) and **SQL** to model customer attributes, lifecycle stages, and purchase behaviour; translated segment definitions into marketing targeting rules and **campaign audience selection** logic for promotional and loyalty programme deployment.
- Tracked financial, market and **customer impact of implemented incentives and loyalty plans** by sourcing **EPOS and third-party customer data**, validating data accuracy across multiple sources, and calculating campaign ROI and customer lifetime value uplift **by segment and category**, presenting findings and corrective actions to senior stakeholders.
- Established **management by exception** reporting frameworks and **KPI benchmarks** for future customer growth and value expansion using **DAX measures, Power Query ETL**, and **data warehousing** principles; monitored root causes of performance variance and advised on priority improvements through statistical analysis and trend identification.
- Supported change-control and **UAT coordination** across interconnected business systems and loyalty platforms; collaborated with business analysts, marketing teams, and operations stakeholders to gather requirements, validate data flows for campaign execution, and deliver post-implementation triage ensuring **data accuracy** and system reliability in production loyalty environments.

SKILLS

Customer Analytics & CRM – customer segmentation, cohort analysis, retention analysis, customer value modelling, CRM performance tracking, loyalty programme analysis, campaign ROI calculation, customer lifetime value.

Data Analysis & Insight Tools – Power BI, Tableau, DAX, Power Query, data visualisation, advanced segmentation, statistical analysis, hypothesis testing, trend identification, management by exception reporting.

Data & Databases – SQL, PostgreSQL, Oracle, MySQL, EPOS data integration, third-party data sources, data warehousing, data modelling, ETL pipeline development, data validation, data accuracy techniques.

Programming & Data Science – Python, pandas, NumPy, scikit-learn, data transformation, exploratory data analysis, supervised and unsupervised learning, regression analysis.

Business Systems & Change Management – ERP integration support, cross-system data flows, business requirement gathering, UAT coordination, change-control management, stakeholder collaboration, post-implementation support.

Domain & Compliance – retail category analysis, customer insight and analysis, marketing analytics, incentive programme performance, GDPR and data protection, customer data governance.

Professional Skills – KPI definition and tracking, business insight presentation, cross-functional collaboration, senior stakeholder communication, data literacy, problem-solving.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021

PROJECTS

Customer Loyalty & Campaign ROI Analytics Platform (HCL Technologies)

- Designed and deployed an integrated **Power BI** and **Tableau** analytics platform aggregating **EPOS transaction data, customer behavioural attributes, and third-party loyalty datasets** to track **campaign uplifts, customer retention, and incentive ROI** across 12+ product categories and 50+ customer segments in real-time.
- Built **advanced customer segmentation engine** using **Python (scikit-learn, pandas)** and **SQL** to classify customers by lifecycle stage, purchase frequency, category affinity, and value tier; created segment-level targeting rules and **audience selection logic** enabling marketing teams to deploy personalised promotions and loyalty offers with **35% improvement in campaign response rate**.
- Engineered **DAX measures and Power Query ETL workflows** to calculate **customer lifetime value, segment-level KPIs, and return on investment benchmarks** from multiple source systems; automated monthly **management by exception** dashboards surfacing root causes of performance variance and flagging priority reactivation and retention opportunities by segment.
- Delivered executive-level dashboards and **KPI tracking frameworks** visualising customer growth targets, loyalty plan financial impact, and category-level value uplift; presented findings and corrective actions to C-suite and category leadership, resulting in **£2.4M incremental customer value** through optimised incentive allocation.
- Established **data validation and accuracy governance** protocols across customer data pipelines, EPOS integrations, and loyalty platform feeds; implemented automated data quality checks reducing downstream reporting errors by **40%** and ensuring reliable performance tracking for all stakeholder reporting.

Customer Segmentation & Retention Analytics Initiative (HCL Technologies)

- Conducted comprehensive **customer behaviour analysis** and **cohort analysis** across 18 months of transactional data using **SQL, Python (pandas, NumPy)**, and statistical methods to identify retention risk segments, high-value dormant cohorts, and reactivation opportunities by category.
- Translated customer segment attributes into **marketing campaign targeting profiles** and promotional audience rules; partnered with CRM and marketing teams to map segments to loyalty programme mechanics, resulting in **28% uplift in customer reactivation** within 90 days of targeted outreach.
- Defined and tracked **segment-level KPI benchmarks** for customer acquisition, retention rate, and value growth using **Power BI** and **Tableau** dashboards; monitored performance against targets and identified root causes of underperformance through data-driven corrective action planning with business stakeholders.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023

- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)